

WASP-43b

R-0500

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_230313 · created 2026-07-20T15:03:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460016.8047 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.011
Transit depth (Rp/Rs)^2	2.32 +/- 0.33 [%]
Orbital Inclination (inc)	83.3 +/- 1.22
Airmass coefficient 1 (a1)	1.0112 +/- 0.0083
Airmass coefficient 2 (a2)	-0.005 +/- 0.012
Scatter in the residuals of the lightcurve fit is	0.82 %
Best Comparison Star	#2 - [66, 351]
Optimal Aperture	3.64
Optimal Annulus	5.96
Transit Duration (day)	0.0537 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.152 $\pm$ 0.011 vs 0.1594 (deviation 0.52 σ)
Duration fit vs published	0.0537 d vs 0.0512 d (deviation 0.62 σ)
Mid-time O-C	-0.6 min
Depth SNR	10.8
Residual scatter	0.82 %

Light curve

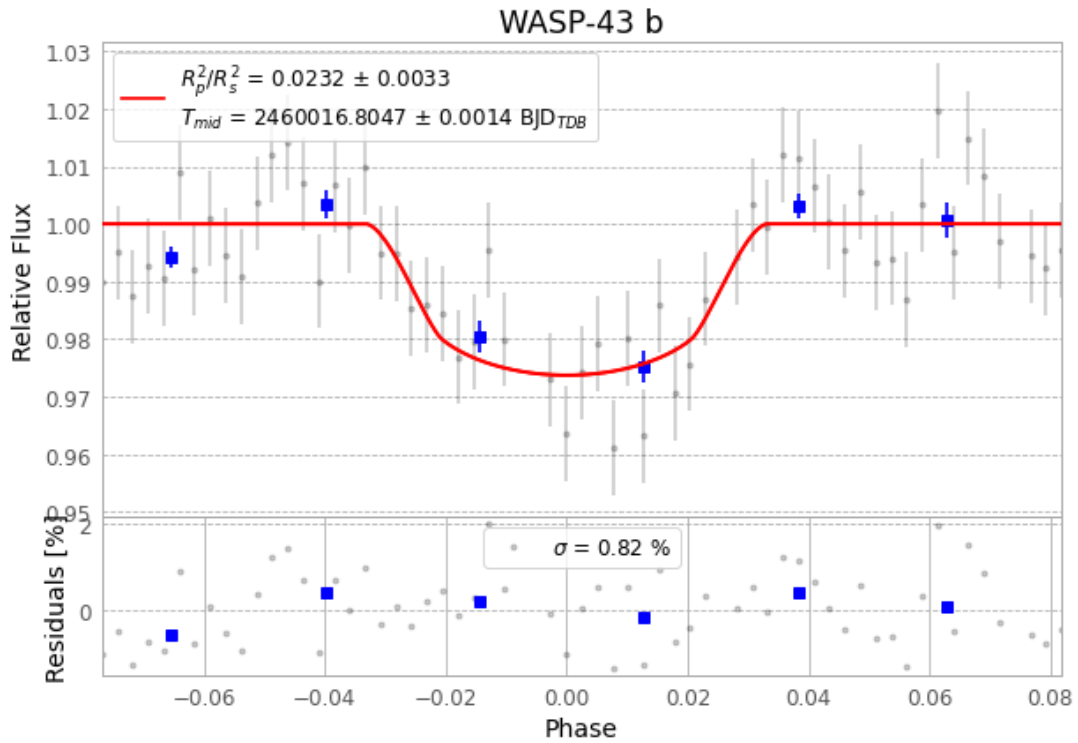


Figure 1 — FinalLightCurve_WASP-43 b_2023-03-12.png (SHA-256 32eb8bde3f19e71b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [445, 180], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c41ad9cb7f8030d5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

61 FITS frames (40 MB), WASP-43230313053915.FITS → WASP-43230313084515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230313.log (1f5a529d87f5...), FinalLightCurve_WASP-43 b_2023-03-12.png (32eb8bde3f19...), FinalLightCurve_WASP-43 b_2023-03-12.csv (e71db498f0dc...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 15:03Z.